校园竞赛获奖证书链上存证与验证系统（基于 FISCO BCOS）证书链存证 · 项目管理实践项目

定位：以“校园竞赛获奖证书”为窄场景，在 FISCO BCOS 上实现证书哈希的存证与验证，并产出项目章程、需求规格、WBS、风险登记册、测试记录、复盘等全生命周期项目管理文档。

1. 安装 openssl 和 curl，FISCO BCOS 的建链脚本依赖这两个工具

```
sudo apt update
```

```
sudo apt install -y openssl curl
```

```
[sudo] cyx 的密码：
命中:1 http://mirrors.aliyun.com/ubuntu focal InRelease
获取:2 http://mirrors.aliyun.com/ubuntu focal-security InRelease [128 kB]
获取:3 http://mirrors.aliyun.com/ubuntu focal-updates InRelease [128 kB]
获取:4 http://mirrors.aliyun.com/ubuntu focal-backports InRelease [128 kB]
获取:5 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 DEP-11 Metadata [74.7 kB]
获取:6 http://mirrors.aliyun.com/ubuntu focal-security/universe amd64 DEP-11 Metadata [160 kB]
获取:7 http://mirrors.aliyun.com/ubuntu focal-security/multiverse amd64 DEP-11 Metadata [940 B]
获取:8 http://mirrors.aliyun.com/ubuntu focal-updates/main amd64 DEP-11 Metadata [275 kB]
获取:9 http://mirrors.aliyun.com/ubuntu focal-updates/universe amd64 DEP-11 Metadata [446 kB]
获取:10 http://mirrors.aliyun.com/ubuntu focal-updates/multiverse amd64 DEP-11 Metadata [940 B]
获取:11 http://mirrors.aliyun.com/ubuntu focal-backports/main amd64 DEP-11 Metadata [8,008 B]
获取:12 http://mirrors.aliyun.com/ubuntu focal-backports/universe amd64 DEP-11 Metadata [30.5 kB]
已下载 1,380 kB，耗时 1秒 (1,007 kB/s)
正在读取软件包列表... 完成
正在分析软件包的依赖关系树
正在读取状态信息... 完成
有 4 个软件包可以升级。请执行‘apt list --upgradable’来查看它们。
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/binary-i386/Packages' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/binary-amd64/Packages' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/i18n/Translation-en' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/i18n/Translation-zh' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/i18n/Translation-zh_CN' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/dep11/Components-amd64.yml' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/dep11/icons-48x48.tar' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/dep11/icons-64x64.tar' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/dep11/icons-64x64@2.tar' 的获取。(sources.list 中的组件名称是否拼写错误？)
W: 鉴于仓库 'http://mirrors.aliyun.com/ubuntu focal-backports InRelease' 不含有 'multiversex' 组件，跳过配置
文件 'multiversex/cnf/Commands-amd64' 的获取。(sources.list 中的组件名称是否拼写错误？)
```

```
cyx@cyx-virtual-machine:~$ sudo apt install -y openssl curl
正在读取软件包列表... 完成
正在分析软件包的依赖关系树
正在读取状态信息... 完成
curl 已经是最新版 (7.68.0-1ubuntu2.25)。
openssl 已经是最新版 (1.1.1f-1ubuntu2.24)。
openssl 已设置为手动安装。
升级了 0 个软件包，新安装了 0 个软件包，要卸载 0 个软件包，有 4 个软件包未被升级。
```

2．创建目录并下载脚本

cd ~

mkdir -p fisco

cd fisco

curl -#LO https://github.com/FISCO-BCOS/FISCO-BCOS/releases/download/v2.11.0/build_chain.sh

chmod u+x build_chain.sh

```
cyx@cyx-virtual-machine:~$ cd ~
cyx@cyx-virtual-machine:~$ mkdir -p fisco
cyx@cyx-virtual-machine:~$ cd fisco
cyx@cyx-virtual-machine:~/fisco$ curl -#LO https://github.com/FISCO-BCOS/FISCO-BCOS/releases/download/v2.11.0
/build_chain.sh
curl: (52) Empty reply from server
```

原因：网络从 GitHub 拉取报错 curl: (52) Empty reply from server 是很常见的网络阻断问题

使用腾讯云镜像重新下载：

curl -#LO https://osp-1257653870.cos.ap-guangzhou.myqcloud.com/FISCO-BCOS/FISCO-BCOS/releases/v2.11.0/build_chain.sh

```
cyx@cyx-virtual-machine:~/fisco$ curl -#LO https://osp-1257653870.cos.ap-guangzhou.myqcloud.com/FISCO-BCOS/FI
SCO-BCOS/releases/v2.11.0/build_chain.sh
######################################################################################################## 100.0%
```

赋权并测试脚本

chmod u+x build_chain.sh

./build_chain.sh -h

```
cyx@cyx-virtual-machine:~/fisco$ chmod u+x build_chain.sh
cyx@cyx-virtual-machine:~/fisco$ ./build_chain.sh -h
./build_chain.sh: 行 1: 未预期的符号“newline”附近有语法错误
./build_chain.sh: 行 1: `<?xml version='1.0' encoding='utf-8' ?>'
cyx@cyx-virtual-machine:~/fisco$
```

原因：报错说明刚才下载下来的不是脚本，而是一个报错的网页（XML文件）

删掉坏文件，从 Gitee 重新下载：

rm -f build_chain.sh

curl -#LO https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/master/tools/build_chain.sh

```
cyx@cyx-virtual-machine:~/fisco$ rm -f build_chain.sh
cyx@cyx-virtual-machine:~/fisco$ curl -#LO https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/master/tools/build_cha
in.sh
######################################################################################################### 100.0%
```

再此赋权并验证

```
cyx@cyx-virtual-machine:~/fisco$ chmod u+x build_chain.sh
cyx@cyx-virtual-machine:~/fisco$ ./build_chain.sh -h
./build_chain.sh: 行 1: [session-ea4a0798]: 未找到命令
cyx@cyx-virtual-machine:~/fisco$ s
```

原因：报错 行 1: [session-ea4a0798]: 未找到命令 说明你还是下到了一个拦截网页

删掉坏文件，伪装浏览器重新下载：

rm -f build_chain.sh

curl -#LO -A "Mozilla/5.0" https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/master/tools/build_chain.sh

```
cyx@cyx-virtual-machine:~/fisco$ curl -#LO -A "Mozilla/5.0" https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/maste
r/tools/build_chain.sh
######################################################################################################### 100.0%
```

检查：

head -n 2 build_chain.sh

```
cyx@cyx-virtual-machine:~/fisco$ head -n 2 build_chain.sh
[session-1246ed81] Repository or file not foundcyx@cyx-virtual-machine:~/fisco$
```

原因：提示说明 Gitee 的 master 分支确实没有这个文件了（FISCO BCOS 仓库做了调整）

删掉坏文件，用 wget 从指定版本下载：

rm -f build_chain.sh

wget https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/v2.11.0/tools/build_chain.sh

```
cyx@cyx-virtual-machine:~/fisco$ wget https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/v2.11.0/tools/build_chain.s
h
--2026-11-07 22:39:53--  https://gitee.com/FISCO-BCOS/FISCO-BCOS/raw/v2.11.0/tools/build_chain.sh
正在解析主机 gitee.com (gitee.com)... 180.76.198.225, 180.76.199.13, 180.76.198.77
正在连接 gitee.com (gitee.com)|180.76.198.225|:443... 已连接。
已发出 HTTP 请求，正在等待回应... 302 Found
位置: https://raw.giteeusercontent.com/FISCO-BCOS/FISCO-BCOS/raw/v2.11.0/tools/build_chain.sh?signature=sX9fl
XBmF0iY8ve-i8IwFxfPjOKS36ln76a5bjgYA8whuoWqlCWIB06SKv4ofJqAKCFx4WweYy8VsdVXtLsGgOK9p-jacfhJO_q-_P5PpREeL6Fy8e
MDcXTfRnT6wZ_llijZWNHmA1-er5h4AMvq2VStl4qBfAjBu1dSg5_o9UQNOSoDImr3QibglvOqnRILVZ3bMgvAgLhlaakAtA6-ruVJmAqoOck
qL8mUFY1AVglumv-56Qq0iKuJWiZT7_NUCPN77i-EKIKUZp5qa75oOiiLAaaZ8Fj2SzADGj_6yGEO2mR_ [跟随至新的 URL]
--2026-11-07 22:39:53--  https://raw.giteeusercontent.com/FISCO-BCOS/FISCO-BCOS/raw/v2.11.0/tools/build_chain
.sh?signature=sX9flXBmF0iY8ve-i8IwFxfPjOKS36ln76a5bjgYA8whuoWqlCWIB06SKv4ofJqAKCFx4WweYy8VsdVXtLsGgOK9p-jacfh
JO_q-_P5PpREeL6Fy8eMDcXTfRnT6wZ_llijZWNHmA1-er5h4AMvq2VStl4qBfAjBu1dSg5_o9UQNOSoDImr3QibglvOqnRILVZ3bMgvAgLhl
aakAtA6-ruVJmAqoOckqL8mUFY1AVglumv-56Qq0iKuJWiZT7_NUCPN77i-EKIKUZp5qa75oOiiLAaaZ8Fj2SzADGj_6yGEO2mR_
正在解析主机 raw.giteeusercontent.com (raw.giteeusercontent.com)... 180.76.199.86
正在连接 raw.giteeusercontent.com (raw.giteeusercontent.com)|180.76.199.86|:443... 已连接。
已发出 HTTP 请求，正在等待回应... 200 OK
长度： 未指定 [text/plain]
正在保存至: “build_chain.sh”

build_chain.sh                [  <=>                                  ]  70.77K   229KB/s    用时 0.3s

2026-11-07 22:39:54 (229 KB/s) - “build_chain.sh” 已保存 [72465]
```

检查文件并赋权

```
cyx@cyx-virtual-machine:~/fisco$ head -n 2 build_chain.sh
#!/bin/bash

cyx@cyx-virtual-machine:~/fisco$ chmod u+x build_chain.sh
```

建链：

bash build_chain.sh -l 127.0.0.1:4 -p 30300,20200,8545

```
cyx@cyx-virtual-machine:~/fisco$ bash build_chain.sh -l 127.0.0.1:4 -p 30300,20200,8545
[INFO] Downloading fisco-bcos binary from https://github.com/FISCO-BCOS/FISCO-BCOS/releases/download/v2.11.0/
fisco-bcos.tar.gz ...
######################################################################## 100.0%
=============================================================
Generating CA key...
=============================================================
Generating keys and certificates ...
Processing IP=127.0.0.1 Total=4 Agency=agency Groups=1
=============================================================
Generating configuration files ...
Processing IP=127.0.0.1 Total=4 Agency=agency Groups=1
=============================================================
[INFO] Start Port      : 30300 20200 8545
[INFO] Server IP       : 127.0.0.1:4
[INFO] Output Dir      : /home/cyx/fisco/nodes
[INFO] CA Path         : /home/cyx/fisco/nodes/cert/
[INFO] RSA channel     : true
=============================================================
[INFO] Execute the download_console.sh script in directory named by IP to get FISCO-BCOS console.
e.g.  bash /home/cyx/fisco/nodes/127.0.0.1/download_console.sh -f
=============================================================
[INFO] All completed. Files in /home/cyx/fisco/nodes
```

启动所有节点

bash nodes/127.0.0.1/start_all.sh

```
cyx@cyx-virtual-machine:~/fisco$ bash nodes/127.0.0.1/start_all.sh
try to start node0
try to start node1
try to start node2
try to start node3
 node0 start successfully
 node2 start successfully
 node1 start successfully
 node3 start successfully
```

检查节点进程是否在运行

ps -ef | grep fisco-bcos

```
cyx@cyx-virtual-machine:~/fisco$ ps -ef | grep fisco-bcos
cyx         3276    1455  1 23:09 pts/0    00:00:00 /home/cyx/fisco/nodes/127.0.0.1/node0/../fisco-bcos -c config.ini
cyx         3277    1455  1 23:09 pts/0    00:00:00 /home/cyx/fisco/nodes/127.0.0.1/node1/../fisco-bcos -c config.ini
cyx         3278    1455  1 23:09 pts/0    00:00:00 /home/cyx/fisco/nodes/127.0.0.1/node2/../fisco-bcos -c config.ini
cyx         3282    1455  1 23:09 pts/0    00:00:00 /home/cyx/fisco/nodes/127.0.0.1/node3/../fisco-bcos -c config.ini
cyx         3705    2028  0 23:10 pts/0    00:00:00 grep --color=auto fisco-bcos
```

2. 安装控制台（Console）

安装 Java 环境（控制台依赖）

sudo apt install -y default-jdk

```
cyx@cyx-virtual-machine:~$ sudo apt install -y default-jdk
[sudo] cyx 的密码：
正在读取软件包列表... 完成
正在分析软件包的依赖关系树
正在读取状态信息... 完成
将会同时安装下列软件：
  ca-certificates-java default-jdk-headless default-jre default-jre-headless
  fonts-dejavu-extra java-common libatk-wrapper-java libatk-wrapper-java-jni
  libice-dev libpthread-stubs0-dev libsm-dev libx11-dev libxau-dev libxcb1-dev
  libxdmcp-dev libxt-dev openjdk-11-jdk openjdk-11-jdk-headless openjdk-11-jre
  openjdk-11-jre-headless x11proto-core-dev x11proto-dev xorg-sgml-doctools
  xtrans-dev
建议安装：
  libice-doc libsm-doc libx11-doc libxcb-doc libxt-doc openjdk-11-demo
  openjdk-11-source visualvm fonts-ipafont-gothic fonts-ipafont-mincho
  fonts-wqy-microhei | fonts-wqy-zenhei
下列【新】软件包将被安装：
  ca-certificates-java default-jdk default-jdk-headless default-jre
  default-jre-headless fonts-dejavu-extra java-common libatk-wrapper-java
  libatk-wrapper-java-jni libice-dev libpthread-stubs0-dev libsm-dev
  libx11-dev libxau-dev libxcb1-dev libxdmcp-dev libxt-dev openjdk-11-jdk
  openjdk-11-jdk-headless openjdk-11-jre openjdk-11-jre-headless
  x11proto-core-dev x11proto-dev xorg-sgml-doctools xtrans-dev
升级了 0 个软件包，新安装了 25 个软件包，要卸载 0 个软件包，有 3 个软件包未被升级。
需要下载 119 MB 的归档。
解压缩后会消耗 277 MB 的额外空间。
获取:1 http://mirrors.aliyun.com/ubuntu focal/main amd64 java-common all 0.72 [6,816 B]
获取:2 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 openjdk-11-jre-headless amd64 11.0.27+6~us1
-0ubuntu1~20.04 [38.3 MB]
获取:3 http://mirrors.aliyun.com/ubuntu focal/main amd64 default-jre-headless amd64 2:1.11-72 [3,192 B]
获取:4 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 ca-certificates-java all 20190405ubuntu1.1
[12.4 kB]
获取:5 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 openjdk-11-jre amd64 11.0.27+6~us1-0ubuntu1
~20.04 [194 kB]
获取:6 http://mirrors.aliyun.com/ubuntu focal/main amd64 default-jre amd64 2:1.11-72 [1,084 B]
获取:7 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 openjdk-11-jdk-headless amd64 11.0.27+6~us1
-0ubuntu1~20.04 [73.8 MB]
获取:8 http://mirrors.aliyun.com/ubuntu focal/main amd64 default-jdk-headless amd64 2:1.11-72 [1,140 B]
获取:9 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 openjdk-11-jdk amd64 11.0.27+6~us1-0ubuntu1
~20.04 [2,876 kB]
获取:10 http://mirrors.aliyun.com/ubuntu focal/main amd64 default-jdk amd64 2:1.11-72 [1,096 B]
获取:11 http://mirrors.aliyun.com/ubuntu focal/main amd64 fonts-dejavu-extra all 2.37-1 [1,953 kB]
获取:12 http://mirrors.aliyun.com/ubuntu focal/main amd64 libatk-wrapper-java all 0.37.1-1 [53.0 kB]
获取:13 http://mirrors.aliyun.com/ubuntu focal/main amd64 libatk-wrapper-java-jni amd64 0.37.1-1 [45.1 kB]
获取:14 http://mirrors.aliyun.com/ubuntu focal/main amd64 xorg-sgml-doctools all 1:1.11-1 [12.9 kB]
获取:15 http://mirrors.aliyun.com/ubuntu focal/main amd64 x11proto-dev all 2019.2-1ubuntu1 [594 kB]
获取:16 http://mirrors.aliyun.com/ubuntu focal/main amd64 x11proto-core-dev all 2019.2-1ubuntu1 [2,620 B]
获取:17 http://mirrors.aliyun.com/ubuntu focal/main amd64 libice-dev amd64 2:1.0.10-0ubuntu1 [47.8 kB]
获取:18 http://mirrors.aliyun.com/ubuntu focal/main amd64 libpthread-stubs0-dev amd64 0.4-1 [5,384 B]
获取:19 http://mirrors.aliyun.com/ubuntu focal/main amd64 libsm-dev amd64 2:1.2.3-1 [17.0 kB]
获取:20 http://mirrors.aliyun.com/ubuntu focal/main amd64 libxau-dev amd64 1:1.0.9-0ubuntu1 [9,552 B]
获取:21 http://mirrors.aliyun.com/ubuntu focal/main amd64 libxdmcp-dev amd64 1:1.1.3-0ubuntu1 [25.3 kB]
获取:22 http://mirrors.aliyun.com/ubuntu focal/main amd64 xtrans-dev all 1.4.0-1 [68.9 kB]
获取:23 http://mirrors.aliyun.com/ubuntu focal/main amd64 libxcb1-dev amd64 1.14-2 [80.5 kB]
获取:24 http://mirrors.aliyun.com/ubuntu focal-security/main amd64 libx11-dev amd64 2:1.6.9-2ubuntu1.6 [648 k
B]
获取:25 http://mirrors.aliyun.com/ubuntu focal/main amd64 libxt-dev amd64 1:1.1.5-1 [395 kB]
已下载 119 MB，耗时 1分 45秒 (1,139 kB/s)
正在选中未选择的软件包 java-common。
(正在读取数据库 ... 系统当前共安装有 186861 个文件和目录。)
准备解压 .../00-java-common_0.72_all.deb  ...
正在解压 java-common (0.72) ...
正在选中未选择的软件包 openjdk-11-jre-headless:amd64。
准备解压 .../01-openjdk-11-jre-headless_11.0.27+6~us1-0ubuntu1~20.04_amd64.deb  ...
正在解压 openjdk-11-jre-headless:amd64 (11.0.27+6~us1-0ubuntu1~20.04) ...
正在选中未选择的软件包 default-jre-headless。
准备解压 .../02-default-jre-headless_2%3a1.11-72_amd64.deb  ...
正在解压 default-jre-headless (2:1.11-72) ...
正在选中未选择的软件包 ca-certificates-java。
准备解压 .../03-ca-certificates-java_20190405ubuntu1.1_all.deb  ...
正在解压 ca-certificates-java (20190405ubuntu1.1) ...
```

```
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/java 来在自动模式中提供 /usr/bin/java (java)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jjs 来在自动模式中提供 /usr/bin/jjs (jjs)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/keytool 来在自动模式中提供 /usr/bin/keytool
(keytool)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/rmid 来在自动模式中提供 /usr/bin/rmid (rmid)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/rmiregistry 来在自动模式中提供 /usr/bin/rmir
egistry (rmiregistry)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/pack200 来在自动模式中提供 /usr/bin/pack200
(pack200)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/unpack200 来在自动模式中提供 /usr/bin/unpack
200 (unpack200)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/lib/jexec 来在自动模式中提供 /usr/bin/jexec (jex
ec)
正在设置 openjdk-11-jre:amd64 (11.0.27+6~us1-0ubuntu1~20.04) ...
正在设置 openjdk-11-jdk-headless:amd64 (11.0.27+6~us1-0ubuntu1~20.04) ...
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jar 来在自动模式中提供 /usr/bin/jar (jar)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jarsigner 来在自动模式中提供 /usr/bin/jarsig
ner (jarsigner)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/javac 来在自动模式中提供 /usr/bin/javac (jav
ac)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/javadoc 来在自动模式中提供 /usr/bin/javadoc
(javadoc)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/javap 来在自动模式中提供 /usr/bin/javap (jav
ap)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jcmd 来在自动模式中提供 /usr/bin/jcmd (jcmd)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jdb 来在自动模式中提供 /usr/bin/jdb (jdb)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jdeprscan 来在自动模式中提供 /usr/bin/jdeprs
can (jdeprscan)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jdeps 来在自动模式中提供 /usr/bin/jdeps (jde
ps)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jfr 来在自动模式中提供 /usr/bin/jfr (jfr)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jimage 来在自动模式中提供 /usr/bin/jimage (j
image)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jinfo 来在自动模式中提供 /usr/bin/jinfo (jin
fo)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jlink 来在自动模式中提供 /usr/bin/jlink (jli
nk)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jmap 来在自动模式中提供 /usr/bin/jmap (jmap)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jmod 来在自动模式中提供 /usr/bin/jmod (jmod)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jps 来在自动模式中提供 /usr/bin/jps (jps)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jrunscript 来在自动模式中提供 /usr/bin/jruns
cript (jrunscript)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jshell 来在自动模式中提供 /usr/bin/jshell (j
shell)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jstack 来在自动模式中提供 /usr/bin/jstack (j
stack)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jstat 来在自动模式中提供 /usr/bin/jstat (jst
at)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jstatd 来在自动模式中提供 /usr/bin/jstatd (j
statd)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/rmic 来在自动模式中提供 /usr/bin/rmic (rmic)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/serialver 来在自动模式中提供 /usr/bin/serial
ver (serialver)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jaotc 来在自动模式中提供 /usr/bin/jaotc (jao
tc)
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jhsdb 来在自动模式中提供 /usr/bin/jhsdb (jhs
db)
正在设置 default-jre (2:1.11-72) ...
正在设置 libxdmcp-dev:amd64 (1:1.1.3-0ubuntu1) ...
正在设置 default-jdk-headless (2:1.11-72) ...
正在设置 x11proto-core-dev (2019.2-1ubuntu1) ...
正在设置 openjdk-11-jdk:amd64 (11.0.27+6~us1-0ubuntu1~20.04) ...
update-alternatives: 使用 /usr/lib/jvm/java-11-openjdk-amd64/bin/jconsole 来在自动模式中提供 /usr/bin/jconsol
e (jconsole)
正在设置 libxcb1-dev:amd64 (1.14-2) ...
正在设置 libx11-dev:amd64 (2:1.6.9-2ubuntu1.6) ...
正在设置 default-jdk (2:1.11-72) ...
正在设置 libxt-dev:amd64 (1:1.1.5-1) ...
cyx@cyx-virtual-machine:~$
```

运行官方自带的控制台下载脚本

bash nodes/127.0.0.1/download_console.sh

```
cyx@cyx-virtual-machine:~/fisco$ bash nodes/127.0.0.1/download_console.sh
Downloading console  from https://github.com/FISCO-BCOS/console/releases/download/v2.11.0/console.tar.gz
######################################################################################################## 100.0%

gzip: stdin: not in gzip format
tar: Child returned status 1
tar: Error is not recoverable: exiting now
```

原因：GitHub的老毛病了，下载下来的根本不是压缩包，而是一个网页拦截提示

从国内腾讯云镜像下载控制台：

wget https://osp-1257653870.cos.ap-guangzhou.myqcloud.com/FISCO-BCOS/console/releases/v2.11.0/console.tar.gz

```
cyx@cyx-virtual-machine:~/fisco$ wget https://osp-1257653870.cos.ap-guangzhou.myqcloud.com/FISCO-BCOS/console
/releases/v2.11.0/console.tar.gz
--2026-11-07 23:20:52--  https://osp-1257653870.cos.ap-guangzhou.myqcloud.com/FISCO-BCOS/console/releases/v2.
11.0/console.tar.gz
正在解析主机 osp-1257653870.cos.ap-guangzhou.myqcloud.com (osp-1257653870.cos.ap-guangzhou.myqcloud.com)... 2
7.155.119.140, 113.240.77.195, 175.6.91.141, ...
正在连接 osp-1257653870.cos.ap-guangzhou.myqcloud.com (osp-1257653870.cos.ap-guangzhou.myqcloud.com)|27.155.1
19.140|:443... 已连接。
已发出 HTTP 请求，正在等待回应... 403 Forbidden
2026-11-07 23:20:52 错误 403: Forbidden。
```

原因：腾讯云COS镜像的直链可能已经过期或被限制了

使用Gitee镜像脚本下载并启动

rm -f console.tar.gz

curl -#LO https://gitee.com/FISCO-BCOS/console/raw/master/tools/download_console.sh

```
cyx@cyx-virtual-machine:~/fisco$ rm -f console.tar.gz
cyx@cyx-virtual-machine:~/fisco$ curl -#LO https://gitee.com/FISCO-BCOS/console/raw/master/tools/download_con
sole.sh
######################################################################################################## 100.0%
 -=O=#    #    #      #
```

运行脚本下载控制台

bash download_console.sh -v 2.11.0

```
cyx@cyx-virtual-machine:~/fisco$ bash download_console.sh -v 2.11.0
[WARN] 2.11.0 is not supported. Please set one of 0.4 0.5 0.6 0.8
```

原因：参数用错了。-v 参数是用来指定 Solidity 编译器版本的（只支持 0.4、0.5、0.6、0.8），而不是控制台版本。要指定控制台版本，需要用 -c 参数。

重新下载并启动控制台

rm -f download_console.sh

从 Gitee 镜像下载正确的脚本（master-2.0 分支）

curl -#LO https://gitee.com/FISCO-BCOS/console/raw/master-2.0/tools/download_console.sh

```
cyx@cyx-virtual-machine:~/fisco$ curl -#LO https://gitee.com/FISCO-BCOS/console/raw/master-2.0/tools/download
_console.sh
######################################################################################################## 100.0%
 -=O#-  #    #     #
```

运行脚本自动下载控制台，从 Gitee 直接下载控制台压缩包

wget

https://gitee.com/FISCO-BCOS/console/releases/download/v2.10.0/console.tar.gz

```
cyx@cyx-virtual-machine:~/fisco$ wget https://gitee.com/FISCO-BCOS/console/releases/download/v2.10.0/console.
tar.gz
--2026-11-07 23:26:49--  https://gitee.com/FISCO-BCOS/console/releases/download/v2.10.0/console.tar.gz
正在解析主机 gitee.com (gitee.com)... 180.76.198.225, 180.76.199.13, 180.76.198.77
正在连接 gitee.com (gitee.com)|180.76.198.225|:443... 已连接。
已发出 HTTP 请求，正在等待回应... 404 Not Found
2026-11-07 23:26:50 错误 404: Not Found。
```

原因： Gitee 的下载链接失效

清理并获取正确的下载脚本

1. **清理之前可能损坏的文件**

rm -f console.tar.gz download_console.sh

2. **获取最新的下载脚本（从Gitee的master-2.0分支）**

curl -#LO

https://gitee.com/FISCO-BCOS/console/raw/master-2.0/tools/download_console.sh

```
cyx@cyx-virtual-machine:~/fisco$ rm -f console.tar.gz download_console.sh
cyx@cyx-virtual-machine:~/fisco$ curl -#LO https://gitee.com/FISCO-BCOS/console/raw/master-2.0/tools/download
_console.sh
######################################################################################################## 100.0%
 -=O#-  #    #     #
cyx@cyx-virtual-machine:~/fisco$
```

运行脚本，自动下载控制台

给脚本添加执行权限

chmod u+x download_console.sh

运行脚本，自动下载控制台

bash download_console.sh

```
cyx@cyx-virtual-machine:~/fisco$ bash download_console.sh
[INFO] Downloading console 2.10.0 from https://github.com/FISCO-BCOS/console/releases/download/v2.10.0/consol
e.tar.gz
#                                                                                                        1.1%
```

```
cyx@cyx-virtual-machine:~/fisco$ bash download_console.sh
[INFO] Downloading console 2.10.0 from https://github.com/FISCO-BCOS/console/releases/download/v2.10.0/consol
e.tar.gz
##########################################################################                               75.4%
curl: (18) transfer closed with 13674698 bytes remaining to read

[WARN] Download console failed, please switch to better network and try again!

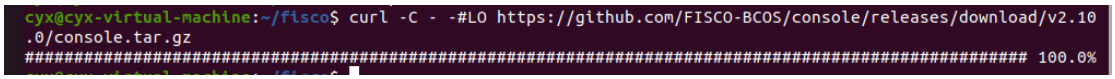
gzip: stdin: unexpected end of file
tar: 归档文件中异常的 EOF
tar: 归档文件中异常的 EOF
tar: Error is not recoverable: exiting now
[WARN] unzip console failed, please try again!
```

原因：GitHub下载大文件时网络中断

断点续传

使用 -C - 参数继续下载

curl -C - -#LO https://github.com/FISCO-BCOS/console/releases/download/v2.10.0/console.tar.gz



解压控制台

tar -zxvf console.tar.gz

```
console/start.sh
console/sol2java.sh
console/conf/
console/conf/config-example.toml
console/conf/log4j2.xml
console/conf/group-generate-config.toml
console/get_gm_account.sh
console/lib/
console/lib/netty-transport-sctp-4.1.100.Final.jar
console/lib/netty-transport-4.1.100.Final.jar
console/lib/tcnative-2.0.51.0.jar
console/lib/netty-codec-4.1.100.Final.jar
console/lib/log4j-api-2.19.0.jar
console/lib/netty-codec-dns-4.1.100.Final.jar
console/lib/toml4j-0.7.2.jar
console/lib/netty-common-4.1.100.Final.jar
console/lib/commons-cli-1.3.1.jar
console/lib/netty-transport-native-unix-common-4.1.100.Final.jar
console/lib/webank-blockchain-hsm-crypto-1.0.0-GMT0018.jar
console/lib/netty-codec-mqtt-4.1.100.Final.jar
console/lib/netty-codec-socks-4.1.100.Final.jar
console/lib/webank-blockchain-java-crypto-1.0.3.jar
console/lib/commons-io-2.7.jar
console/lib/netty-all-4.1.100.Final.jar
console/lib/netty-transport-native-epoll-4.1.100.Final-linux-aarch_64.jar
console/lib/netty-transport-classes-epoll-4.1.100.Final.jar
console/lib/jackson-databind-2.15.4.jar
console/lib/jline-3.25.0.jar
console/lib/netty-codec-stomp-4.1.100.Final.jar
console/lib/lombok-1.18.6.jar
console/lib/netty-handler-4.1.100.Final.jar
console/lib/log4j-slf4j-impl-2.19.0.jar
console/lib/netty-resolver-dns-classes-macos-4.1.100.Final.jar
console/lib/slf4j-api-1.7.36.jar
console/lib/netty-codec-memcache-4.1.100.Final.jar
console/lib/bcos-code-generator-1.4.0.jar
console/lib/javapoet-1.13.0.jar
console/lib/netty-transport-native-kqueue-4.1.100.Final-osx-aarch_64.jar
console/lib/jackson-core-2.15.4.jar
console/lib/netty-transport-udt-4.1.100.Final.jar
console/lib/netty-handler-proxy-4.1.100.Final.jar
console/lib/commons-lang3-3.14.0.jar
console/lib/netty-resolver-dns-4.1.100.Final.jar
console/lib/netty-transport-native-kqueue-4.1.100.Final-osx-x86_64.jar
console/lib/gson-2.10.1.jar
console/lib/solcJ-0.4.25.2.jar
console/lib/console-table-builder-1.2.jar
console/lib/netty-transport-classes-kqueue-4.1.100.Final.jar
console/lib/netty-codec-smtp-4.1.100.Final.jar
console/lib/netty-codec-haproxy-4.1.100.Final.jar
console/lib/bcprov-jdk18on-1.77.jar
console/lib/log4j-web-2.19.0.jar
console/lib/netty-codec-http-4.1.100.Final.jar
console/lib/netty-codec-redis-4.1.100.Final.jar
console/lib/log4j-core-2.19.0.jar
console/lib/netty-handler-ssl-ocsp-4.1.100.Final.jar
console/lib/netty-resolver-dns-native-macos-4.1.100.Final-osx-aarch_64.jar
console/lib/netty-codec-xml-4.1.100.Final.jar
console/lib/netty-resolver-dns-native-macos-4.1.100.Final-osx-x86_64.jar
console/lib/netty-sm-ssl-context-1.5.2.jar
console/lib/netty-codec-http2-4.1.100.Final.jar
console/lib/jackson-annotations-2.15.4.jar
console/lib/netty-resolver-4.1.100.Final.jar
console/lib/fisco-bcos-java-sdk-2.10.0.jar
console/lib/netty-transport-native-epoll-4.1.100.Final-linux-x86_64.jar
console/lib/netty-buffer-4.1.100.Final.jar
console/lib/netty-transport-rxtx-4.1.100.Final.jar
console/lib/picocli-4.6.1.jar
console/lib/jsqlparser-2.0.jar
console/get_account.sh
cyx@cyx-virtual-machine:~/fisco$
```

拷贝节点证书

```
cp -r nodes/127.0.0.1/sdk/* console/conf/
```

```
cyx@cyx-virtual-machine:~/fisco$ cp -r nodes/127.0.0.1/sdk/* console/conf/
```

拷贝配置文件

cp -n console/conf/config-example.toml console/conf/config.toml

```
cyx@cyx-virtual-machine:~/fisco$ cp -n console/conf/config-example.toml console/conf/config.toml
```

启动控制台

cd console

bash start.sh

```
cyx@cyx-virtual-machine:~/fisco/console$ bash start.sh
create BcosSDK failed, error info: init channel network error!
org.fisco.bcos.sdk.network.NetworkException:
* TRACE INFORMATION:
---------------------------
====> STEP1: try to connect nodes with ecdsa context...
<==== STEP1-1: Load certificates for ecdsa context success...
<==== connect nodes failed, reason:
Failed to connect to all the nodes!
* connect to 127.0.0.1:20200 failed! Please make sure the nodes have been started, and the network between th
e SDK and the nodes are connected normally.reason: 拒绝连接: /127.0.0.1:20200

* connect to 127.0.0.1:20201 failed! Please make sure the nodes have been started, and the network between th
e SDK and the nodes are connected normally.reason: 拒绝连接: /127.0.0.1:20201
```

原因：报错的核心是 Failed to connect to all the nodes!，这意味着控制台（SDK）无法连接到区块链节点的 20200 端口

确认节点是否在运行

ps -ef | grep fisco-bcos（新开终端）

原因：之前启动的4个区块链节点不还在运行

重新启动控制台

cd console（检查有没有在console文件中，没有则使用）

bash start.sh

```
cyx@cyx-virtual-machine:~/fisco/console$ bash start.sh
=====================================================================================================
Welcome to FISCO BCOS console(2.10.0)!
Type 'help' or 'h' for help. Type 'quit' or 'q' to quit console.
 _______ ______  ______   ______   ______       _______   ______   ______   ______
|       \/      \ /      \ /      \ /      \     |       \ /      \ /      \ /      \
| $$$$$$$\$$$$$$|  $$$$$$|  $$$$$$|  $$$$$$\    | $$$$$$$|  $$$$$$|  $$$$$$|  $$$$$$\
| $$__     | $$ | $$___\$| $$   \$| $$  | $$    | $$__/ $| $$   \$| $$  | $| $$___\$$
| $$  \    | $$  \$$    \| $$      | $$  | $$    | $$    $| $$      | $$  | $$\$$    \
| $$$$$    | $$  _\$$$$$$| $$   __| $$  | $$    | $$$$$$$| $$   __| $$  | $$_\$$$$$$\
| $$      _| $$_|  \__| $| $$__/  | $$__/ $$    | $$__/ $| $$__/  | $$__/ $|  \__| $$
| $$     |   $$ \\$$    $$\$$    $$\$$    $$    | $$    $$\$$    $$\$$    $$\$$    $$
 \$$      \$$$$$$ \$$$$$$  \$$$$$$  \$$$$$$      \$$$$$$$  \$$$$$$  \$$$$$$  \$$$$$$

=====================================================================================================
[group:1]>
```

3. 部署存证智能合约

新建存证合约文件

进入存证合约目录

```
cd contracts/solidity
```

新建一个名为 CertificateEvidence.sol 的文件

```
nano CertificateEvidence.sol
```

输入以下代码：

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.4.25;

contract CertificateEvidence {
    // 定义存证的数据结构
    struct Evidence {
        address owner;      // 存证人的地址
        uint256 timestamp;  // 存证时间
        bool exists;        // 是否存在
    }

    // 用文件哈希值映射到存证信息
    mapping(bytes32 => Evidence) private evidences;

    // 存证功能：输入文件哈希，记录在链上
    function createEvidence(bytes32 hash) public {
        require(!evidences[hash].exists, "This evidence already exists!");
        evidences[hash] = Evidence(msg.sender, block.timestamp, true);
    }
```

```
    // 验证功能：输入文件哈希，返回存证信息
    function verifyEvidence(bytes32 hash) public view returns (address, uint256, bool) {
        Evidence memory e = evidences[hash];
        return (e.owner, e.timestamp, e.exists);
    }
}
```

```
cyx@cyx-virtual-machine:~/fisco/console$ # 进入存证合约目录
cyx@cyx-virtual-machine:~/fisco/console$ cd contracts/solidity
cyx@cyx-virtual-machine:~/fisco/console/contracts/solidity$
cyx@cyx-virtual-machine:~/fisco/console/contracts/solidity$ # 新建一个名为 CertificateEvidence.sol 的文件
```

部署合约

cd ~/fisco/console

bash start.sh

```
cyx@cyx-virtual-machine:~/fisco/console$ bash start.sh
=====================================================================================================
Welcome to FISCO BCOS console(2.10.0)!
Type 'help' or 'h' for help. Type 'quit' or 'q' to quit console.
 ________ ______  ______   ______  ______       _______  ______  ______  ______
|        |      \/      \ /      \/      \     |       \/      \/      \/      \
| $$$$$$$$\$$$$$|  $$$$$$|  $$$$$$|  $$$$$$\    | $$$$$$$|  $$$$$$|  $$$$$$|  $$$$$$\
| $$__     | $$ | $$___\$| $$   \$| $$  | $$    | $$__/ $| $$   \$| $$  | $| $$___\$$
| $$  \    | $$  \$$    \| $$     | $$  | $$    | $$    $| $$     | $$  | $$\$$    \
| $$$$$    | $$  _\$$$$$$| $$   __| $$  | $$    | $$$$$$$| $$   __| $$  | $$_\$$$$$$\
| $$      _| $$_|  \__| $| $$__/  | $$__/ $$    | $$__/ $| $$__/  | $$__/ $|  \__| $$
| $$     |   $$ \\$$    $$\$$    $$\$$    $$    | $$    $$\$$    $$\$$    $$\$$    $$
 \$$      \$$$$$$ \$$$$$$  \$$$$$$  \$$$$$$      \$$$$$$$  \$$$$$$  \$$$$$$  \$$$$$$

=====================================================================================================
[group:1]>
```

（再次看到 [group1]: 后），输入以下命令部署合约：

deploy CertificateEvidence.sol

```
[group:1]> deploy CertificateEvidence.sol
transaction hash: 0x6fcbeee9cc2379539e3bb78317844ee7866054a6ea2024687272d721afacf97a
contract address: 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5
currentAccount: 0x5bff9e15fdaee1c79058f41dbc2238b1eeb38b69
```

存证（把证书哈希写进区块链）

call CertificateEvidence 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5 createEvidence 0x1111111111111111111111111111111111111111111111111111111111111111

```
[group:1]> call CertificateEvidence 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5 createEvidence 0x1111111111111
1111111111111111111111111111111111111111111111111
transaction hash: 0xa77a4792c93d2d24bb1ae713c92b7c236f5d474568f46098238c059dd0499f7c
---------------------------------------------------------------------------------------------
transaction status: 0x0
description: transaction executed successfully
---------------------------------------------------------------------------------------------
Transaction inputs:
Input value size:1
Input types: (BYTES)
Input values:(hex://0x1111111111111111111111111111111111111111111111111111111111111111)
---------------------------------------------------------------------------------------------
Receipt message: Success
Return message: Success
Return values:[]
---------------------------------------------------------------------------------------------
Event logs
Event: {}
```

验证刚才存入的证书（应该返回 true）

call CertificateEvidence 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5 verifyEvidence 0x1111111111111111111111111111111111111111111111111111111111111111

```
[group:1]> call CertificateEvidence 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5 verifyEvidence 0x1111111111111
1111111111111111111111111111111111111111111111111
---------------------------------------------------------------------------------------------
Return code: 0
description: transaction executed successfully
Return message: Success
---------------------------------------------------------------------------------------------
Return value size:3
Return types: (ADDRESS, UINT, BOOL)
Return values:(0x5bff9e15fdaee1c79058f41dbc2238b1eeb38b69, 1794073098856, true)
---------------------------------------------------------------------------------------------
```

若没有证书，则应该返回false

call CertificateEvidence 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5 verifyEvidence 0x2222222222222222222222222222222222222222222222222222222222222222

```
[group:1]> call CertificateEvidence 0xe90a32b9d1abe00013c2bc447446aa6fe2daf8a5 verifyEvidence 0x2222222222222
2222222222222222222222222222222222222222222222222
---------------------------------------------------------------------------------------------
Return code: 0
description: transaction executed successfully
Return message: Success
---------------------------------------------------------------------------------------------
Return value size:3
Return types: (ADDRESS, UINT, BOOL)
Return values:(0x0000000000000000000000000000000000000000, 0, false)
---------------------------------------------------------------------------------------------
```

作品集文件夹

建立作品集目录结构

cd ~

mkdir -p fisco-pm-portfolio/docs

mkdir -p fisco-pm-portfolio/evidence-screenshots

cd fisco-pm-portfolio

```
cyx@cyx-virtual-machine:~/fisco/console$ cd ~
cyx@cyx-virtual-machine:~$ mkdir -p fisco-pm-portfolio/docs
cyx@cyx-virtual-machine:~$ mkdir -p fisco-pm-portfolio/evidence-screenshots
cyx@cyx-virtual-machine:~$ cd fisco-pm-portfolio
cyx@cyx-virtual-machine:~/fisco-pm-portfolio$
```

撰写文档《01-项目章程.md》

gedit docs/01-项目章程.md &